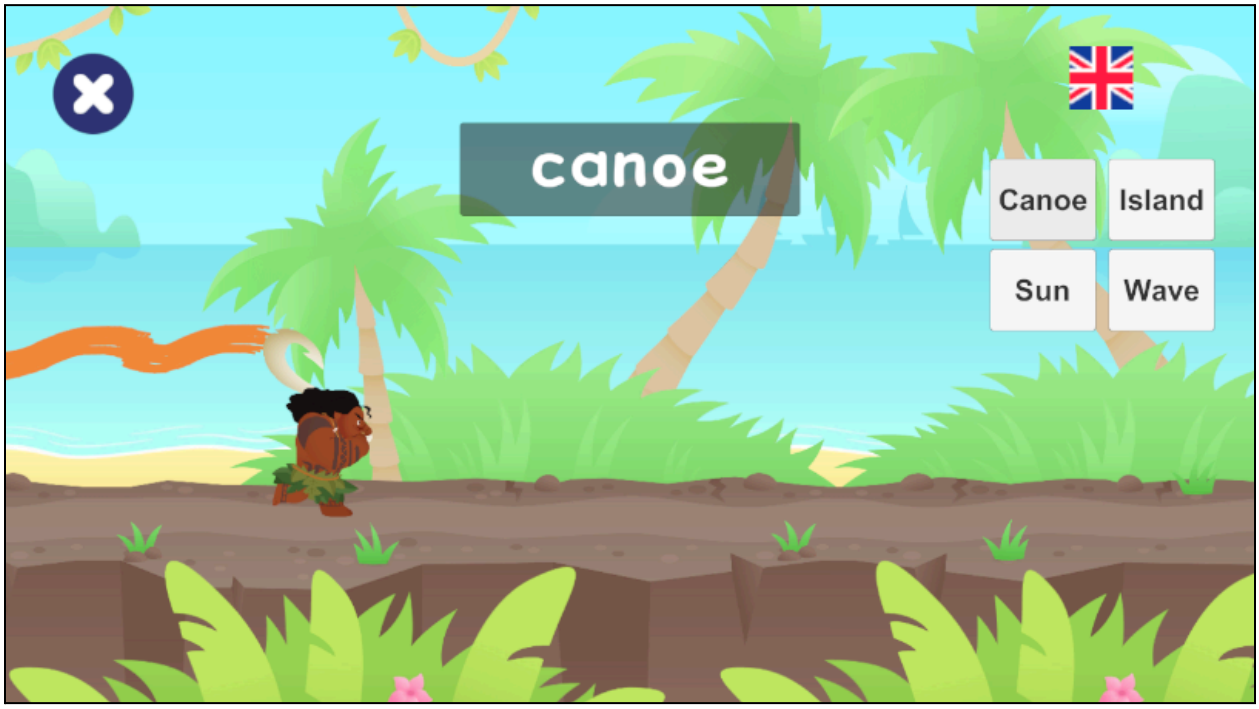
Exercise 2 – *Text Pill Framework* Documentation



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1. Overview

The ***Text Pill Framework*** is a **reusable UI component** designed to **display short messages** (words or sentences) for a **limited duration**. It is intended for use in ***Lingokids*** activities to present vocabulary, feedback, or short instructions in a **clear and non-intrusive way**.

The component focuses on **simplicity**, **readability**, and **smooth transitions**, making it suitable for **young audiences**.

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2. Features

- Displays **short text messages** with smooth fade and vertical motion.
- **Configurable timing** and **animation** parameters.
- Fully integrated with **Unity Localization Package**.
- Supports **runtime content updates.**
- Designed to integrate with **localization workflows**.
- Supports **dynamic global language switching**.
- Layout **adapts to different word lengths** using TextMeshPro auto-size.

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3. Component Structure

```
TextPillFramework (Prefab Root)
    ├── TextCanvas
        ├── TextPill (Controller)
            ├── Background (Image)
            └── Label (TextMeshProUGUI)
```

- *CanvasGroup* controls fade transitions.
- *RectTransform* controls vertical movement.
- *TextMeshProUGUI* ensures readability and scalability.

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4. Inspector Parameters

- ***Seconds Showing***
  Duration the text remains visible before hiding.
- ***Animation Duration***
  Controls the speed of fade and movement transitions.

- ***Hidden Y Offset***
  Vertical offset used to position the element outside the screen before and after animation.
- ***Canvas Group***
  Reference used to control alpha transitions.
- ***Rect Transform***
  Reference used for position-based animation.

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5. Demo Setup

The demo scene (Assets/Project/Scenes/*DemoScene*) includes **multiple buttons** as presets, that assign **different localization** keys at runtime before triggering the Text Pill animation.

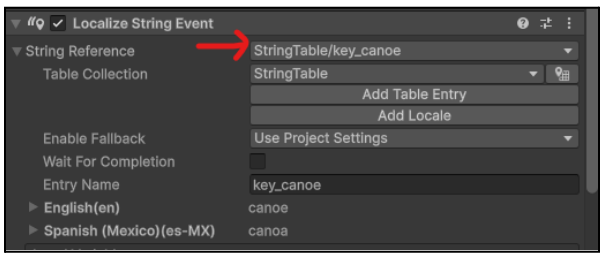
Additionally, the **system supports dynamic language switching**, allowing validation of the component across **different languages during runtime**.

This setup allows **quick validation** of the component with multiple inputs and simulates **real usage in a gameplay context**.

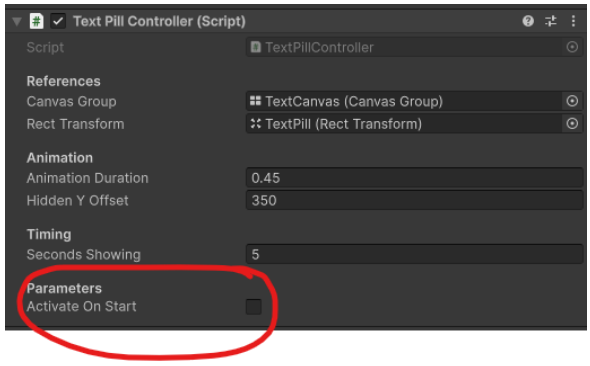
6. Usage

1. Add the *TextPillFramework* prefab to the scene. There are **two methods** of setting the **word to be displayed**, and **two methods** of playing the **animation**.

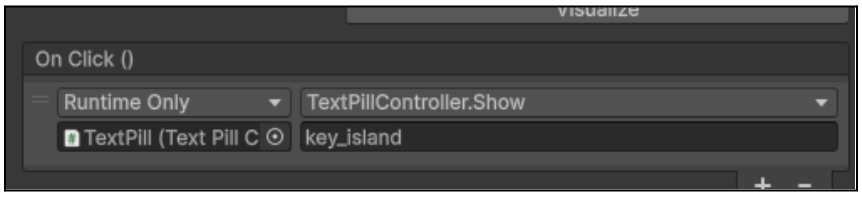
   a. To **change** the **word to be displayed** directly in the **Inspector**, use the *Localize String Event* component, attached to the LabelText gameobject to **select** one of the preset **localization keys**.



   b. Enabling the *ActivateOnStart* parameter on *TextPillComponent* plays the animation when the scene enters in *Play Mode* with the pre-set localization key.



2. The **second method** is to **change** the **localization key** during **runtime**. To do it, reference the ***TextPillController*** Script from a *UI button*.



3. Configure the *OnClick()* event the component using:

   textPillController.Show(*[selected_key]*); where *[selected_key]* can be:

   a. key_island
   b. key_canoe
   c. key_sun
   d. key_wave

4. **Global language switching system** is available through the button on the top-right corner of the *Game View* during *Play Mode*:

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7. Localization Integration

The component is designed to **work with localized** content **without** directly **managing language data**.

Using the ***Localization Package***, a simple localization table has been implemented (English / Mexican Spanish) with a reduced dataset (4 entries) to demonstrate the workflow while keeping the scope concise.

Example:

Key: key_canoe
EN: Canoe
ES_MX: Canoa

At runtime, UI buttons assign different **localization keys** before triggering the Text Pill. This **simulates how the component would receive localized content** dynamically from an **external system**.

The ***Text Pill Framework*** focuses exclusively on **presentation** (timing, animation, readability), while localization **logic remains external**. This **separation** ensures **scalability** and **flexibility** across **multiple languages**.

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8. Design Considerations

- The **timer is not visible**, keeping the UI clean and avoiding unnecessary on-screen elements that could **distract the player**.
- **Smooth easing** is applied to animations to improve **readability** and **comfort**.
- The component **prioritizes clarity** over complexity.
- Text **length variability** is handled using TextMeshPro auto-size, allowing the layout to adapt to **different word lengths**.
- The system is designed to support **dynamic content injection**.

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9. Limitations

- The localization system is intentionally **simplified and limited in scope**, although it **uses** all the **necessary tools of a real product**.

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10. Possible Extensions

- Support for **audio playback** (word pronunciation).
- Additional **visual themes** (success, error, hint).
- Use a **Google Spreadsheet extension** to use an **external database** and enable **simultaneous collaboration**.

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11. Conclusion

The Text Pill component demonstrates a **reusable and scalable UI element** designed for educational gameplay. It separates presentation from content, supports **runtime updates**, and integrates with a **simplified localization workflow**.

The inclusion of a lightweight **localization table** and **runtime key assignment** shows how the component can operate within a **real production pipeline** while maintaining a **focused and concise implementation**.